

A POSITIVE-CHARACTERISTIC COUNTEREXAMPLE TO THE LITERAL Q-CARTIER/BASE-CHANGE EQUIVALENCE

ARGUS OPEN RESEARCH PROJECT

ABSTRACT. AIM Problem 1.1 asks whether the condition that the relative dualizing sheaf of a family of surfaces is $\mathbb{Q}$ -Cartier is equivalent to Kollár's requirement that all reflexive pluricanonical powers commute with arbitrary base change. We verify that the unrestricted equivalence is false. Lee and Nakayama construct a flat projective surface family whose first infinitesimal thickening over $B = \operatorname{Spec} k[t]/(t^2)$, in positive characteristic, is naively or weakly $\mathbb{Q}$ -Gorenstein, while its second reflexive pluricanonical power fails relative S_2 . Their base-change criterion then implies that the map to the reduced closed fiber is not an isomorphism. This gives a direct counterexample to the weak-to-Kollár implication. The example has a nonreduced Artin base and Fano surface fibers; it does not contradict Hacking's characteristic-zero DVR theorem or settle a silently narrowed canonically polarized formulation.

1. THE LITERAL EQUIVALENCE

Let $f : X \rightarrow B$ be a family of surfaces with relative dualizing sheaf $\omega_{X/B}$. For $m > 0$, write

$$\omega_{X/B}^{[m]} = (\omega_{X/B}^{\otimes m})^{\vee\vee}.$$

For a base change $\phi : B' \rightarrow B$, with induced map $\phi_X : X \times_B B' \rightarrow X$, the relevant comparison is

$$(\phi_X^* \omega_{X/B})^{[m]} \longrightarrow \phi_X^* (\omega_{X/B}^{[m]}). \quad (1)$$

AIM Problem 1.1 asks whether the following two conditions are equivalent [4]:

- (i) $\omega_{X/B}$ is $\mathbb{Q}$ -Cartier, equivalently some positive reflexive power is invertible;
- (ii) for every $m > 0$ and every base change $B' \rightarrow B$, the comparison (1) is an isomorphism.

The source calls (ii) Kollár's stronger condition. It places no reduced-base or characteristic-zero assumption in the question itself.

Lee and Nakayama use the following terminology [3, Definition 7.1].

Definition 1.1. For an S_2 morphism with $\mathbb{Q}$ -Gorenstein fibers:

- (a) the morphism is *naively $\mathbb{Q}$ -Gorenstein* if $\omega^{[r]}$ is invertible for some $r > 0$;
- (b) it is *$\mathbb{Q}$ -Gorenstein* if every $\omega^{[m]}$ satisfies relative S_2 .

Their introduction identifies (a) with the weak or Viehweg condition and (b) with the Kollár condition. Along a $\mathbb{Q}$ -Gorenstein fiber, relative S_2 for $\omega^{[m]}$ is equivalent to compatibility with the fiber base-change map [3, Remark 7.6].

2. THE PROJECTIVE SURFACE FAMILY

Lee and Nakayama begin with the contraction of the (-4) -curve on the Hirzebruch surface F_4 , obtaining the weighted projective plane

$$X = \mathbb{P}(1, 1, 4).$$

Their Example 7.4 constructs, over an algebraically closed field k , a flat projective morphism

$$f : Y \longrightarrow T = \mathbb{A}_k^1$$

with the following properties:

- (1) Y is normal and projective;
- (2) the special fiber is $Y_0 \simeq \mathbb{P}(1, 1, 4)$, a $\mathbb{Q}$ -Gorenstein surface of index 2;
- (3) for $t \neq 0$, one has $Y_t \simeq \mathbb{P}^1 \times \mathbb{P}^1$;

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(4) every fiber is $\mathbb{Q}$ -Gorenstein.

The original family over $\mathbb{A}^1$ is not naively $\mathbb{Q}$ -Gorenstein: the canonical self-intersection jumps from $K_{Y_0}^2 = 9$ to $K_{Y_t}^2 = 8$.

Localize T at the origin. Let R be the resulting discrete valuation ring, with maximal ideal $\mathfrak{m}$, and define

$$R_n = R/\mathfrak{m}^{n+1}, \quad T_n = \operatorname{Spec} R_n, \quad Y_n = Y \times_T T_n.$$

In positive residue characteristic, the infinitesimal thickenings behave differently from the whole DVR family.

3. THE ARTIN-THICKENING COUNTEREXAMPLE

Theorem 3.1. *Let k be an algebraically closed field of characteristic $p > 0$. There exists a flat projective family of surfaces*

$$g : Y_1 \longrightarrow B = \operatorname{Spec} k[t]/(t^2)$$

such that g is naively $\mathbb{Q}$ -Gorenstein, but $\omega_{Y_1/B}^{[2]}$ fails relative S_2 . In particular, the closed-point base-change map

$$\omega_{Y_1/B}^{[2]} \otimes_B k \longrightarrow \omega_{\mathbb{P}(1,1,4)/k}^{[2]}$$

is not an isomorphism.

Proof. Apply the preceding construction in characteristic $p > 0$. Since the special fiber is $\mathbb{Q}$ -Gorenstein, Lee–Nakayama Remark 7.28 implies that every infinitesimal thickening $Y_n \rightarrow T_n$ is naively $\mathbb{Q}$ -Gorenstein. For $n = 1$, the base is $B = \operatorname{Spec} k[t]/(t^2)$, so some positive reflexive power of $\omega_{Y_1/B}$ is invertible.

For this same first thickening, Lemma 7.5 of [3] states that $\omega_{Y_1/B}^{[2]}$ is not relatively S_2 and that $Y_1 \rightarrow B$ is not $\mathbb{Q}$ -Gorenstein in the stronger sense. Remark 7.6 identifies this relative- S_2 failure with failure of the displayed base-change map to the closed fiber. The target is invertible because the special fiber has index 2. $\square$

Corollary 3.2. *The unrestricted equivalence in AIM Problem 1.1 is false: weak or naive $\mathbb{Q}$ -Gorensteinness does not imply compatibility of all reflexive pluricanonical powers with arbitrary base change.*

Proof. The family in Theorem 3.1 satisfies condition (i) of the AIM question and fails condition (ii) for $m = 2$ and the base change from B to its reduced closed point. $\square$

4. THE CHARACTERISTIC-ZERO DVR BOUNDARY

The counterexample does not conflict with the special result cited in the AIM problem. Hacking proves the following in characteristic zero [1, Proposition 10.14].

Proposition 4.1 (Hacking). *Let T be the spectrum of a DVR and let $X \rightarrow T$ be a weakly $\mathbb{Q}$ -Gorenstein family of semi-log-canonical surfaces. If the generic fiber is canonical, then the family is $\mathbb{Q}$ -Gorenstein in the base-change sense.*

A smoothing has smooth, hence canonical, generic fiber. Together with the general strong-to-weak implication, this yields equivalence in that restricted setting. The theorem does not cover arbitrary bases, nonreduced Artin thickenings, or positive characteristic.

There is also a bibliographic mismatch in the AIM report. Its bibliography names Hacking’s later paper [2], where the corresponding statement is Lemma 3.4, but the locator “Proposition 10.14” belongs to the earlier thesis [1].

5. SCOPE

Remark 5.1 (Scope). The base $B = \operatorname{Spec} k[t]/(t^2)$ is nonreduced, and the thickened total scheme is not normal. The fibers in the Lee–Nakayama family are of Fano or Del Pezzo type rather than canonically polarized. None of these properties invalidates the counterexample to the literal AIM question, which does not exclude them. Nevertheless, the example is not asserted to settle a modified problem restricted to stable or canonically polarized surfaces over reduced characteristic-zero bases.

The proof above is a source-level reconstruction and requires no numerical computation.

APPENDIX A. CHECKLIST FOR AN INDEPENDENT REFEREE

- (1) Read AIM Problem 1.1 and verify that it requires all $m > 0$ and arbitrary base change without excluding nonreduced bases or positive characteristic.
- (2) Check Lee–Nakayama Definition 7.1 and the weak/naive versus Kollár terminology.
- (3) Verify the fiber descriptions and flat projectivity in Example 7.4.
- (4) Confirm that Remark 7.28 makes the first positive-characteristic Artin thickening naively $\mathbb{Q}$ -Gorenstein.
- (5) Confirm that Lemma 7.5 makes $\omega^{[2]}$ fail relative S_2 on the same thickening.
- (6) Use Remark 7.6 to translate that failure into failure of the closed-fiber base-change map.
- (7) Keep the unrestricted AIM question separate from Hacking’s characteristic-zero DVR theorem and from canonically polarized-only variants.

REFERENCES

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